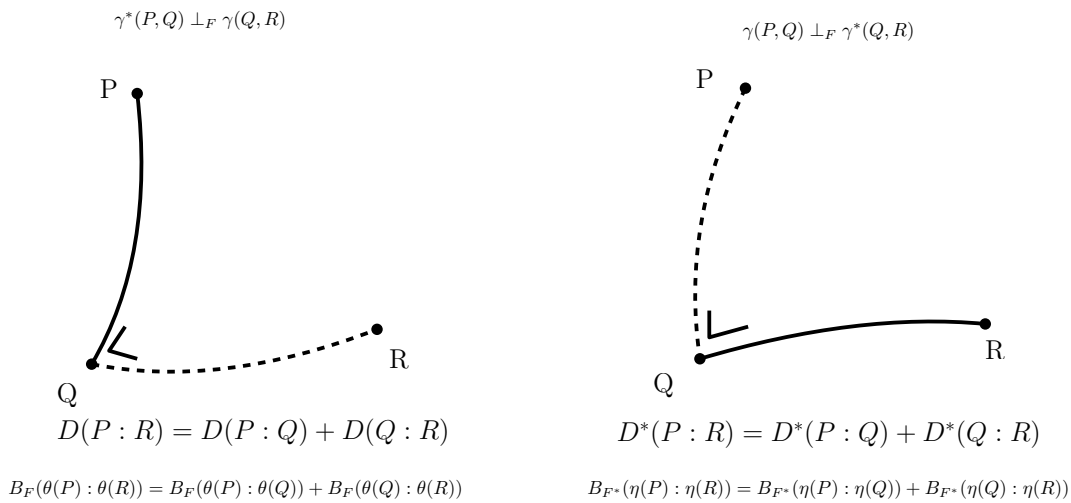

Figure 6: Dual Pythagorean theorems in a dually flat space.

on a Bregman manifold, the primal parallel transport of a vector does not change the contravariant vector components, and the dual parallel transport does not change the covariant vector components. Because the dual connections are flat, the dual parallel transports are path-independent.

Moreover, the dual Pythagorean theorems [76] illustrated in Figure 6 holds. Let $\gamma(P,Q) = \gamma_\nabla(P,Q)$ denote the ∇-geodesic passing through points P and Q, and $\gamma^*(P,Q) = \gamma_{\nabla^*}(P,Q)$ denote the ∇^*-geodesic passing through points P and Q. Two curves γ_1 and γ_2 are orthogonal at point $p = \gamma_1(t_1) = \gamma_2(t_2)$ with respect to the metric tensor g when $g(\dot\gamma_1(t_1), \dot\gamma_2(t_2)) = 0$.

Theorem 6 (Dual Pythagorean identities)**.**

$$\begin{aligned}
\gamma^*(P,Q) \perp \gamma(Q,R) \quad &\Leftrightarrow \quad (\eta(P)-\eta(Q))^\top(\theta(Q)-\theta(R)) \overset{\Sigma}{=} (\eta_i(P)-\eta_i(Q))(\theta_i(Q)-\theta_i(R)) = 0,\\
\gamma(P,Q) \perp \gamma^*(Q,R) \quad &\Leftrightarrow \quad (\theta(P)-\theta(Q))^\top(\eta(Q)-\eta(R)) \overset{\Sigma}{=} (\theta_i(P)-\theta_i(Q))^\top(\eta_i(Q)-\eta_i(R)) = 0.
\end{aligned}$$

We can define dual Bregman projections and characterize when these projections are unique: A *submanifold* $S \subset M$ is said ∇-flat (∇^*-flat) iff. it corresponds to an *affine subspace* in the θ-coordinate system (in the η-coordinate system, respectively).

Theorem 7 (Uniqueness of projections)**.** *The* ∇-projection P_S *of* P *on* S *is* unique *if* S *is* ∇^*-flat *and minimizes the divergence* $D(\theta(P):\theta(Q))$*:*

$$\nabla\text{-}\textit{projection:} \quad P_S = \arg\min_{Q\in S} D(\theta(P):\theta(Q)). \tag{66}$$

The dual ∇^**-projection* P_S^* *is* unique *if* $M \subseteq S$ *is* ∇-flat *and minimizes the divergence* $D(\theta(Q):\theta(P))$*:*

$$\nabla^*\text{-}\textit{projection:} \quad P_S^* = \arg\min_{Q\in S} D(\theta(Q):\theta(P)). \tag{67}$$

Let $S \subset M$ and $S' \subset M$, then we define the divergence between S and S' as

$$D(S:S') := \min_{s\in S, s'\in S'} D(s:s'). \tag{68}$$

When S is a ∇-flat submanifold and S' ∇^*-flat submanifold, the divergence $D(S:S')$ between submanifold S and submanifold S' can be calculated using the method of alternating projections [8]. Let us remark

that Kurose [61] reported a Pythagorean theorem for dually constant curvature manifolds that generalizes the Pythagorean theorems of dually flat spaces.

We shall concisely explain the *space of Bregman spheres* explained in details in [20]. Let D denote the dimension of Θ. We define the lifting of primal coordinates θ to the primal potential function $\mathcal{F} = \{\hat{\theta} = (\theta, \theta_{D+1} = F(\theta)) \ : \ \theta \in \Theta\}$ using an extra dimension θ_{D+1}. A Bregman ball Σ

$$\mathrm{Ball}_F(C : r) := \{P \text{ such that } F(\theta(P)) + F^*(\eta(C)) - \langle \theta(P), \eta(C) \rangle \leq r\} \tag{69}$$

can then be lifted to $\mathcal{F}$: $\hat{\Sigma} = \{\hat{\theta}(P) \ : \ P \in \sigma\}$. The boundary Bregman sphere $\sigma = \partial\Sigma$ is lifted to $\partial\hat{\Sigma} = \hat{\sigma}$, and the lifted points are all supported by a supporting $(D+1)$-dimensional hyperplane (of dimension D):

$$H_{\hat{\sigma}} : \theta_{D+1} = \langle \theta - \theta(C), \eta(C) \rangle + F(\theta(C)) + r. \tag{70}$$

Let $H_{\hat{\sigma}}^-$ denotes the halfspaces bounded by $H_{\hat{\sigma}}$ and containing $\hat{\theta}(C) = (\theta(C), F(\theta(C)))$. A point P belongs to a Bregman ball Σ iff $\theta(\hat{P}) \in H_{\hat{\sigma}}^-$, see [20]. Reciprocally, a $(D+1)$-dimensional hyperplane $H : \theta_{D+1} = \langle \theta, \eta_a \rangle + b$ cutting the potential function $\mathcal{F}$ yields a Bregman sphere σ_H of center C with $\theta(C) = \nabla F^*(\eta_a)$ and radius $r = \langle \nabla F^*(\eta_a), \eta_a \rangle - F(\theta_a) + b = F^*(\eta_a) + b$, where $\theta_a = \nabla F^*(\eta_a)$. It follows that the intersection of k Bregman balls is a $(D-k)$-dimensional Bregman ball, and that a Bregman sphere can be defined by $D+1$ points in general position since an hyperplane in the augmented space is defined by $D+1$ points. We can test whether a point P belongs to a Bregman ball with bounding Bregman sphere passing through $D+1$ points $P_1, \ldots, P_{D+1}$ or not by checking the sign of a $(D+2) \times (D+2)$ determinant:

$$\mathrm{InBregmanBall}_F(P_1, \ldots, P_{d+1}; P) := \mathrm{sign}\left(\begin{vmatrix} 1 & \ldots & 1 & 1 \\ \theta(P_1) & \ldots & \theta(P_{D+1}) & \theta(P) \\ F(\theta(P_1)) & \ldots & F(\theta(P_{D+1})) & F(\theta(P)) \end{vmatrix} \right). \tag{71}$$

We have:

$$\mathrm{InBregmanBall}_F(P_1, \ldots, P_{d+1}; P) : \begin{cases} = -1 & \Leftrightarrow \quad P \in \mathrm{InBregmanBall}_F^\circ(P_1, \ldots, P_{D+1}; P) \\ = 0 & \Leftrightarrow \quad P \in \partial\mathrm{InBregmanBall}_F(P_1, \ldots, P_{D+1}; P) \\ = +1 & \Leftrightarrow \quad P \notin \mathrm{InBregmanBall}_F(P_1, \ldots, P_{D+1}; P) \end{cases} \tag{72}$$

Similarly, a dual-type Bregman ball Σ^* can be defined by

$$\mathrm{Ball}_F^*(C : r) := \{P \text{ such that } F(\theta(C)) + F^*(\eta(P)) - \langle \theta(C), \eta(P) \rangle \leq r\}, \tag{73}$$

and be lifted to the dual potential function $\mathcal{F}^*$. Notice that $\mathrm{Ball}_F^*(C : r) = \mathrm{Ball}_{F^*}(C : r)$.

In general, we have the following quadrilateral relation for Bregman divergences:

Property 4 (Bregman 4-parameter property [37])**.** *For any four points P_1, P_2, Q_1, Q_2, we have the following identity:*

$$\begin{aligned} B_F(\theta(P_1) : \theta(Q_1)) + B_F(\theta(P_2) : \theta(Q_2)) \qquad & -B_F(\theta(P_1) : \theta(Q_2)) - B_F(\theta(P_2) : \theta(Q_1)) \\ & -(\theta(P_2) - \theta(P_1))^\top (\eta(Q_1) - \eta(Q_2)) = 0. \end{aligned} \tag{74}$$

In summary, to define a dually flat space, we need a convex Bregman generator. When the α-geometries are neither dually flat (eg., Cauchy manifolds [79], we may still build a dually flat structure on the manifold by considering some Bregman generator (eg., Bregman-Tsallis generator for the dually flat Cauchy manifold [79]). The dually flat geometry can be investigated under the wider scope of *Hessian manifolds* [120] which consider *locally* potential functions. In general, a dually flat space can be built from any smooth strictly convex generator F. For example, a dually flat geometry can be built on homogeneous cones with the characteristic function F of the cone [120]. Figure 7 illustrates several common constructions of dually flat spaces.

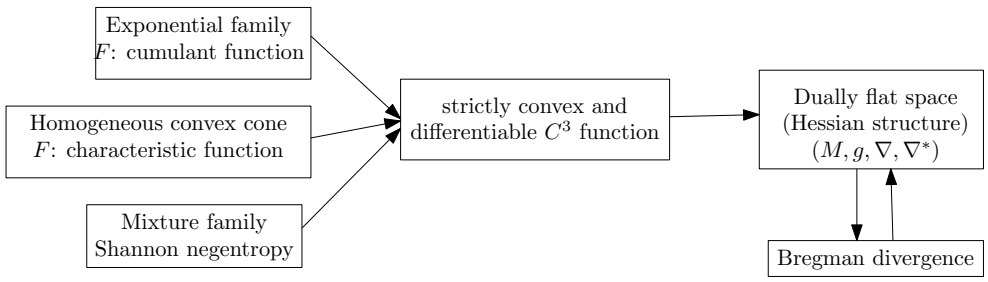

Figure 7: Common dually flat spaces associated to smooth and strictly convex generators.

3.8 Hessian α-geometry: $(M, F, \alpha) \equiv (M, {}^F g, {}^F\nabla^{-\alpha}, {}^F\nabla^{\alpha}$

The dually flat manifold is also called a manifold with a *Hessian structure* [120] induced by a convex potential function F. Since we built two dual affine connections ${}^{B_F}\nabla = {}^F\nabla$ and ${}^{B_F}\nabla^* = {}^F\nabla^* = {}^{F^*}\nabla$, we can build a family of α-geometry as follows:

$$ {}^F g_{ij}(\theta) = \partial_i \partial_j F(\theta), \quad {}^F g^{ij}(\eta) = \partial^i \partial^j F(\eta), \tag{75} $$

and

$$ {}^F\Gamma^{\alpha}_{ijk}(\theta) = \frac{1-\alpha}{2} \partial_i \partial_j \partial_k F(\theta), \quad {}^F\Gamma^{\alpha}_{ijk}{}^*(\eta) = {}^{F^*}\Gamma^{\alpha}_{ijk}(\eta) = \frac{1+\alpha}{2} \partial^i \partial^j \partial^k F^*(\eta). \tag{76} $$

Thus when $\alpha = \pm 1$, the Hessian α-geometry is dually flat.

We now consider information manifolds induced by parametric statistical models.

3.9 Expected α-manifolds of a family of parametric probability distributions: $(\mathcal{P}, {}_{\mathcal{P}}g, {}_{\mathcal{P}}\nabla^{-\alpha}, {}_{\mathcal{P}}\nabla^{\alpha})$

Informally speaking, an *expected manifold* is an information manifold built on a regular parametric family of distributions. It is sometimes called "expected" manifold or "expected" geometry in the literature [140] because the components of the metric tensor g and the Amari-Chentsov cubic tensor C are expressed using statistical expectations $E.[\cdot]$.

Let $\mathcal{P}$ be a *parametric family* of probability distributions:

$$ \mathcal{P} := \{p_\theta(x)\}_{\theta \in \Theta}, \tag{77} $$

with θ belonging to the open parameter space Θ. The order of the family is the dimension of its parameter space. We define the likelihood function[11] $L(\theta; x) := p_\theta(x)$ as a function of θ, and its corresponding *log-likelihood* function:

$$ l(\theta; x) := \log L(\theta; x) = \log p_\theta(x). \tag{78} $$

The *score vector*:

$$ s_\theta = \nabla_\theta l = (\partial_i l)_i, \tag{79} $$

indicates the sensitivity of the likelihood $\partial_i l := : \frac{\partial}{\partial \theta_i} l(\theta; x)$.

The *Fisher information matrix* (FIM) of $D \times D$ for $\dim(\Theta) = D$ is defined by:

$$ {}_{\mathcal{P}}I(\theta) := E_\theta \left[\partial_i l \partial_j l\right]_{ij} \succeq 0, \tag{80} $$

where $\succeq$ denotes the Löwner order. That is, for two symmetric positive-definite matrices A and B, $A \succeq B$ if and only if matrix $A - B$ is positive semidefinite. For regular models [25], the FIM is positive definite: ${}_{\mathcal{P}}I(\theta) \succ 0$, where $A \succ B$ if and only if matrix $A - B$ is positive-definite.

[11] The likelihood function is an *equivalence class* of functions defined modulo a positive scaling factor.